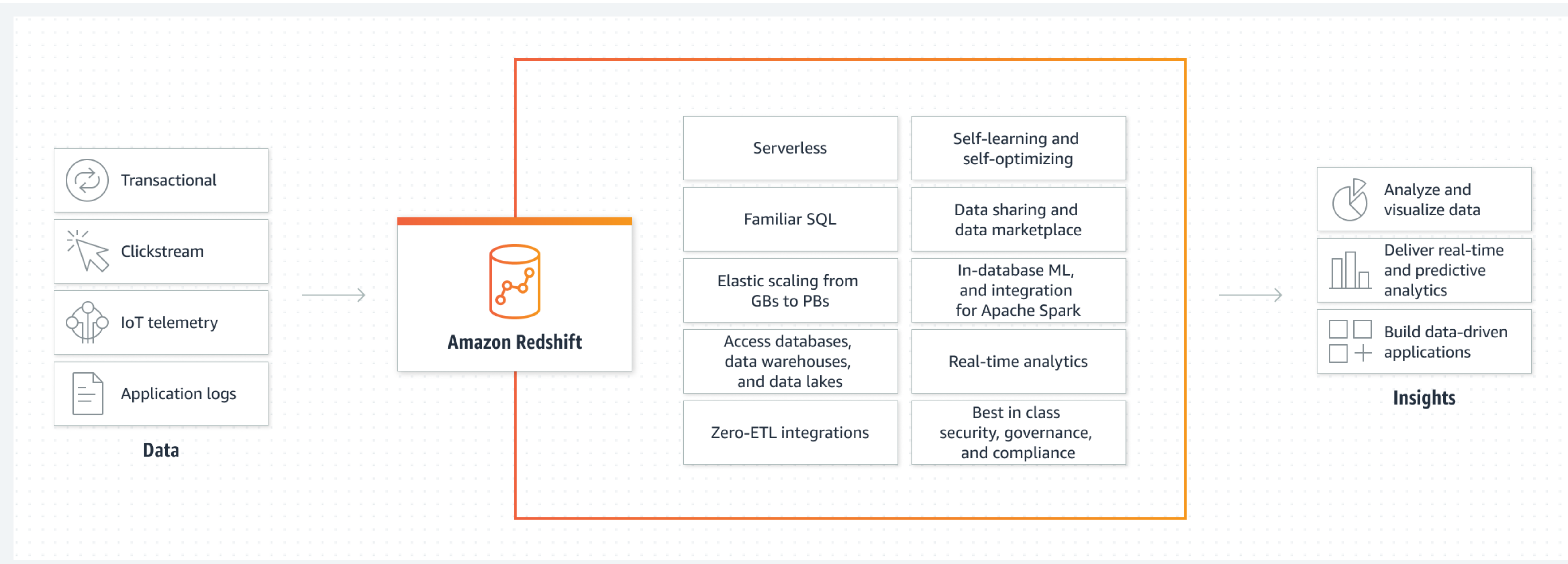
IST — Workshop

Infrastructures pour le stockage et le traitement de données | Academic year 2025/26

IST Workshop

- At the last week of the semester we are going to hold a workshop where groups of students present on a particular topic in data engineering / MLOps.
  - Objective is to go deeper on a concrete cloud service or open source software, based on the theory you saw in class.
- We are going to pretend that you work as a data scientist / data engineer in a Swiss SME and your boss has told you to explore a service / software that appears interesting for the company.
  - Your objective is to find out what is its value for the company, what are its limitations, how much it costs, if there are any alternatives.
  - To be able to go deeper on certain questions, you are going to prepare an **example use case** that demonstrates how the service / software is used.
  - Your boss wants a **critical evaluation** of the service / software. Don't fall for content produced by a marketing department. Distinguish important features from unimportant features. Analyse the service / software as an engineer.
  - At the last week of the semester you are going to **present** your findings in a 15-minute presentation. This presentation includes a **demo** of the use case.

AWS Redshift cloud data warehouse



- Redshift is a data warehouse cloud service
- Based on PostgreSQL
- Can be deployed as dedicated cluster or as serverless service
- Massively Parallel Processing (MPP) SQL engine
- Integration with Apache Spark and other AWS services
- Scales to Petabyes of data

AWS Redshift cloud data warehouse

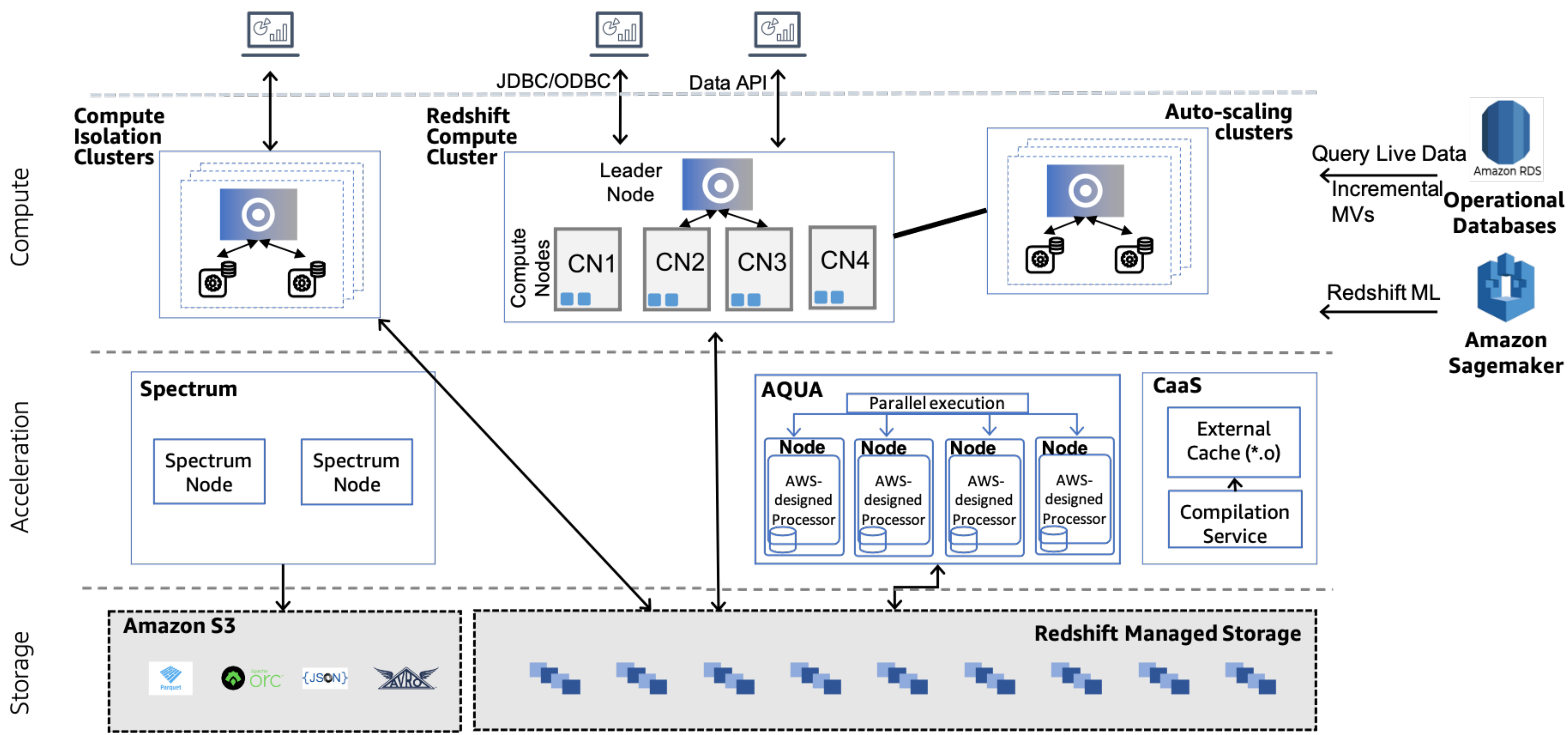
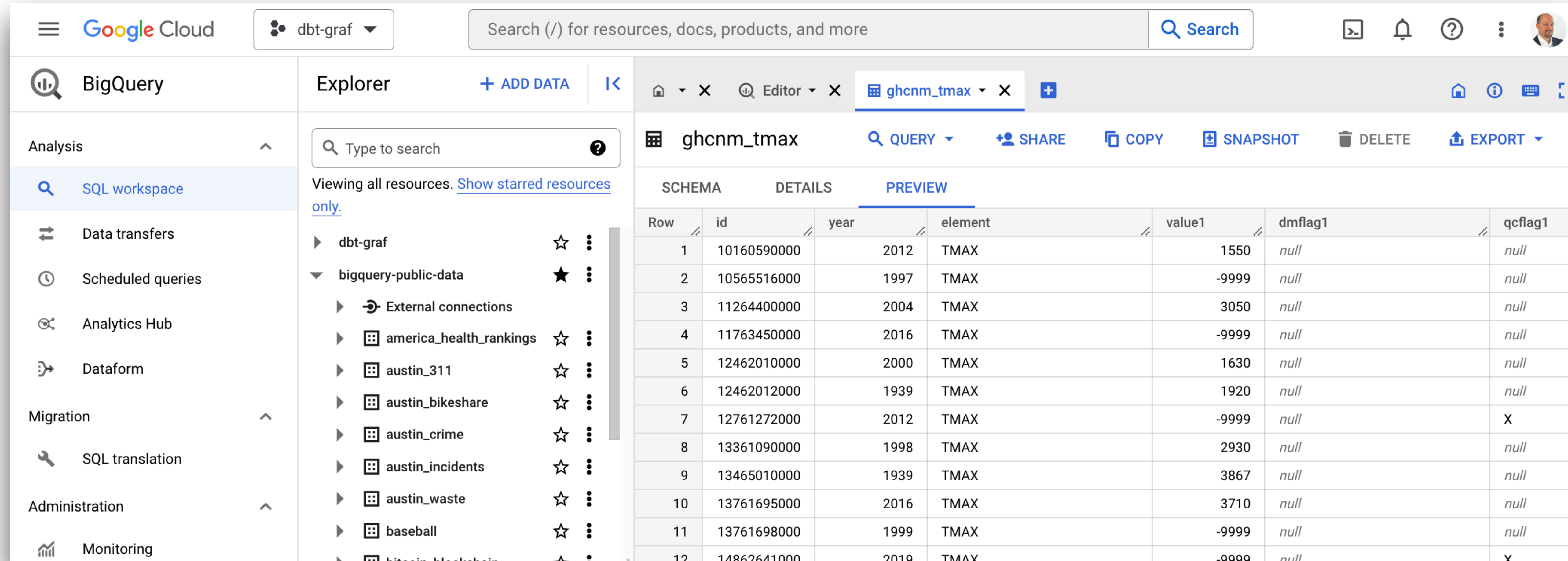


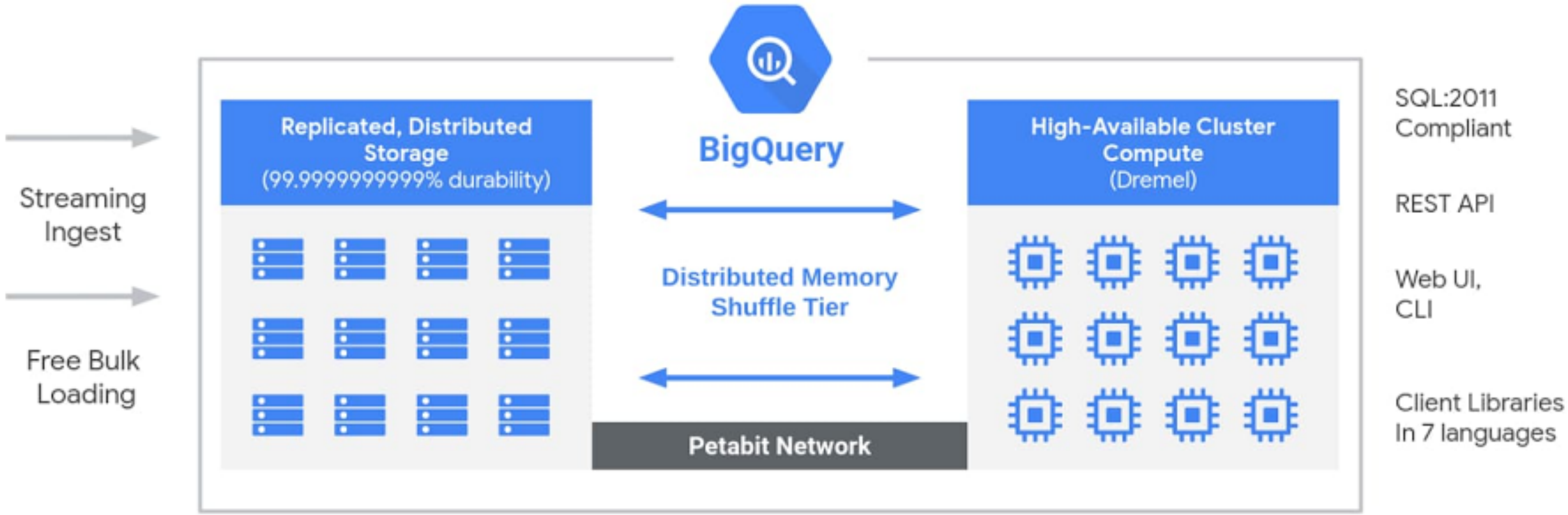
Figure 1: Amazon Redshift Architecture

Google BigQuery cloud data warehouse

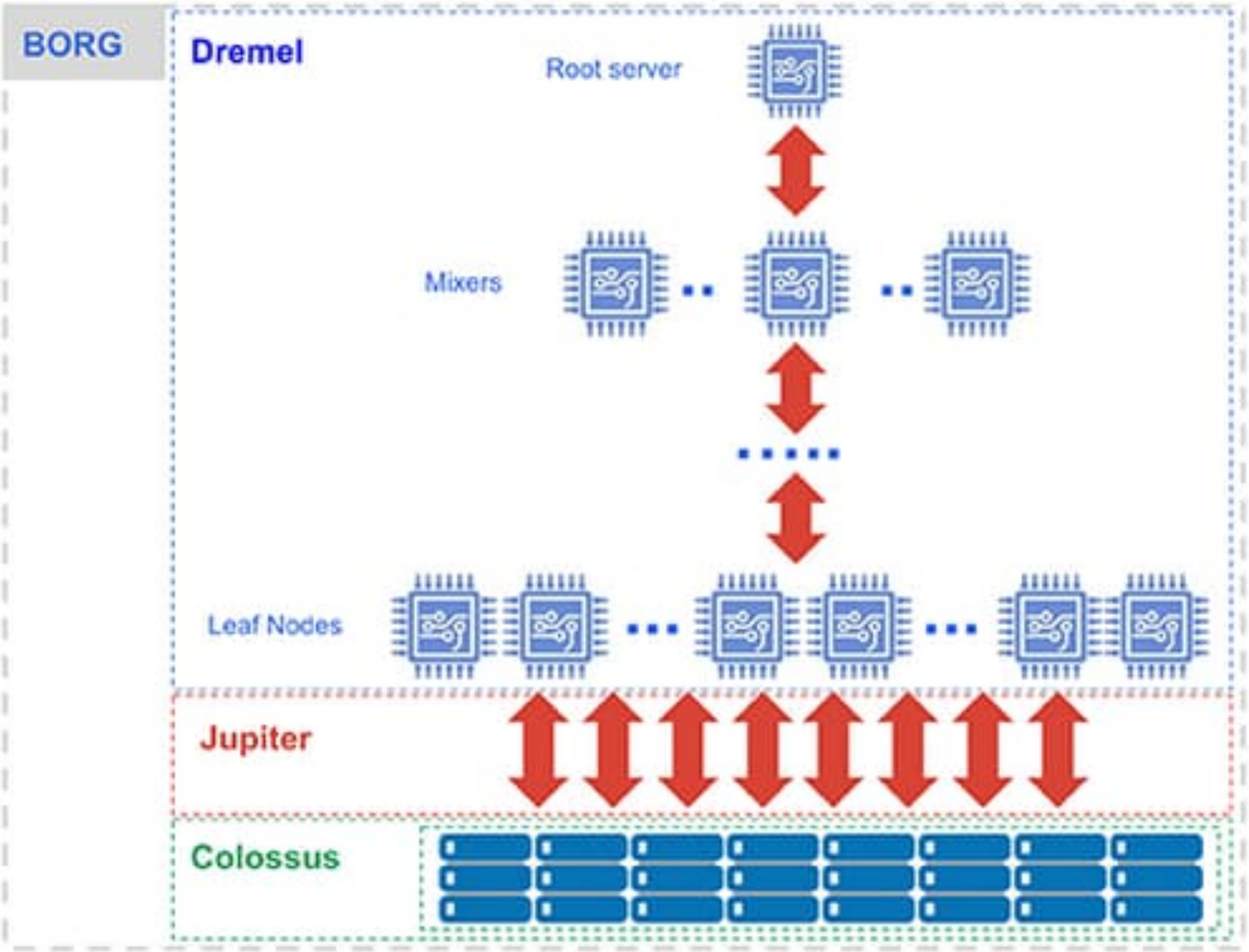


- BigQuery is a data warehouse cloud service
- Based on proprietary Google technologies: Colossus, Jupiter, Dremel and Borg
- Serverless
- Massively Parallel Processing (MPP) SQL engine
- Integration with Apache Spark and other Google Cloud services
- Scales to Petabyes of data

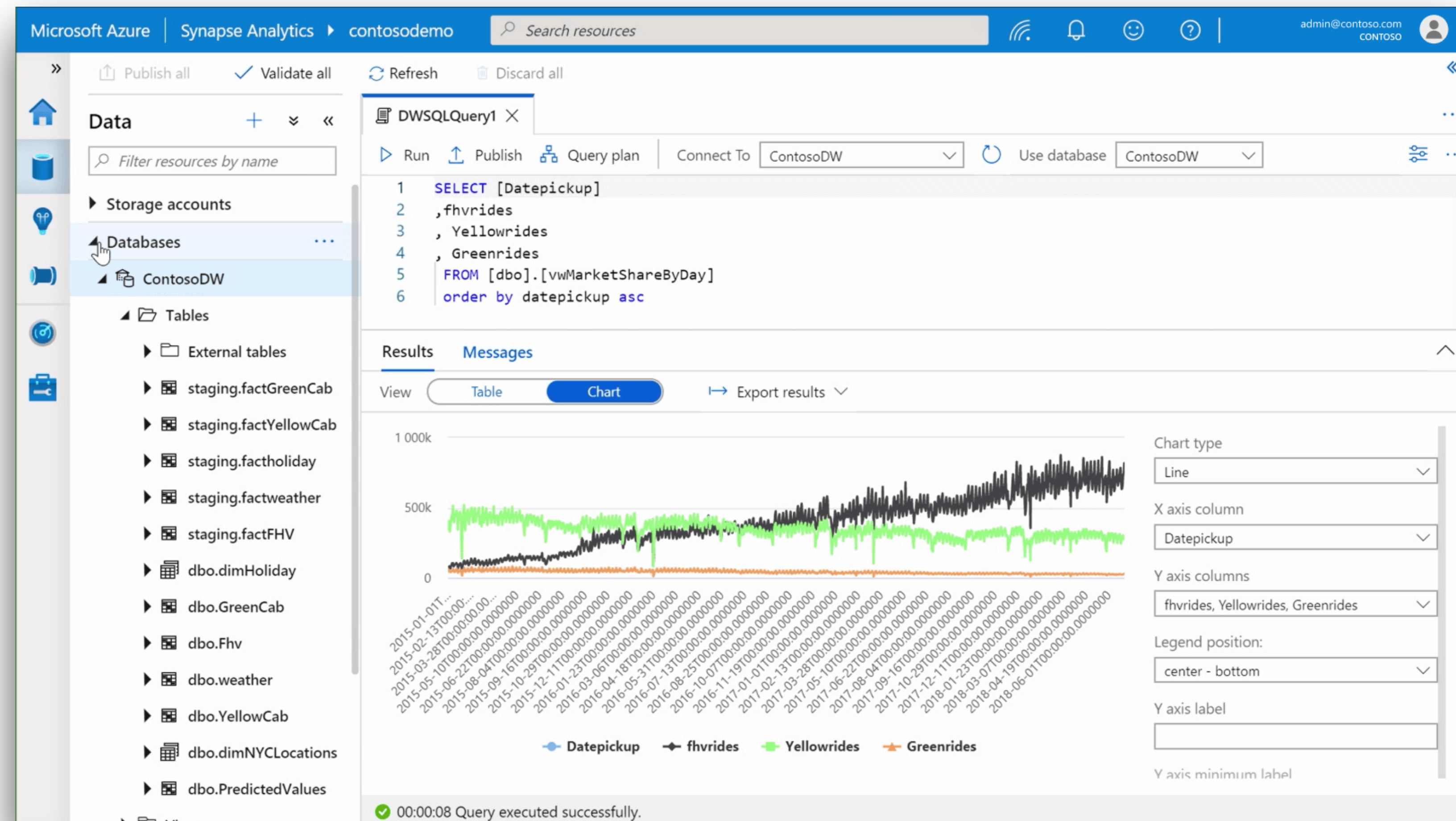
Google BigQuery cloud data warehouse



Google BigQuery cloud data warehouse

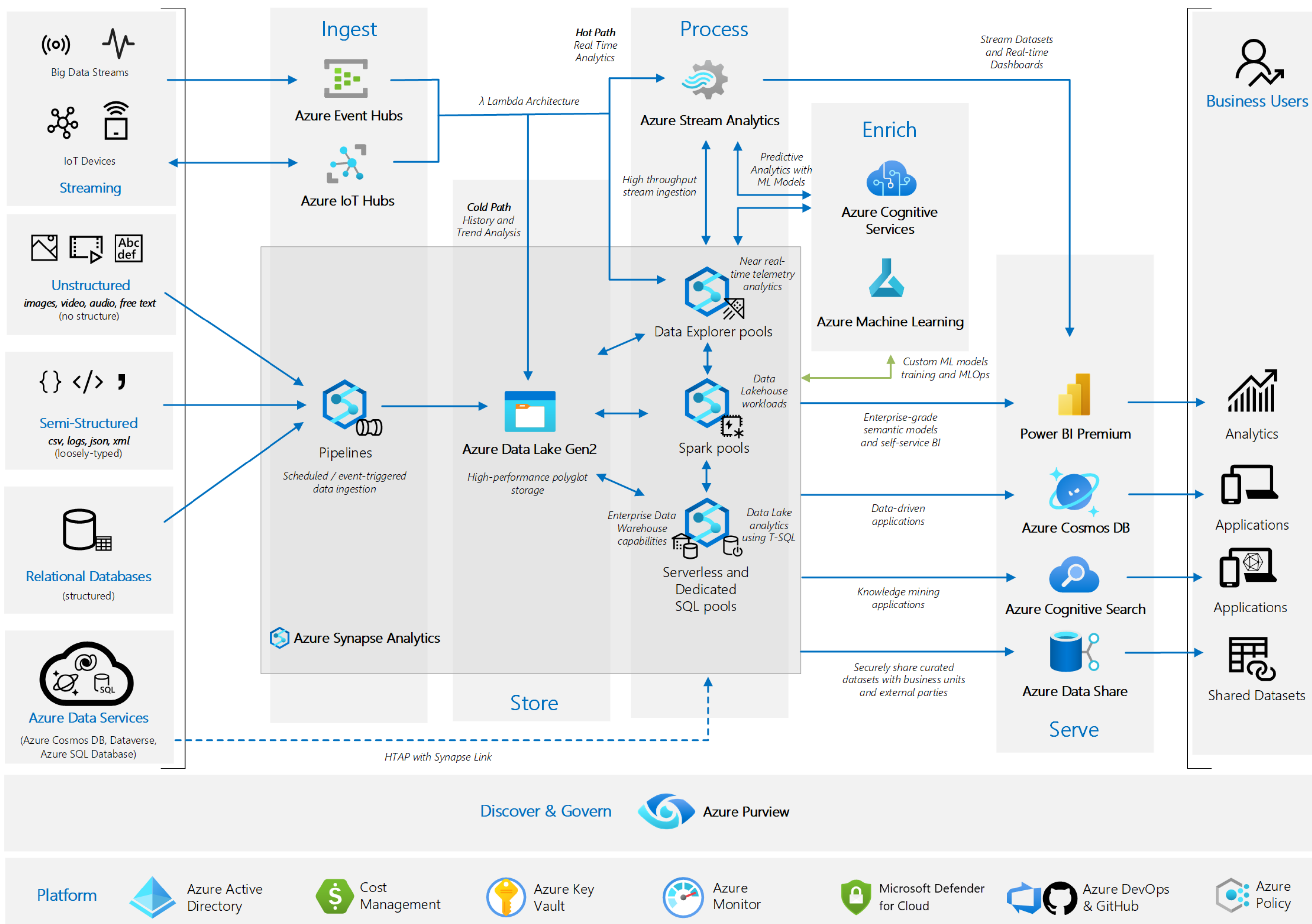


Azure Synapse Analytics cloud data warehouse

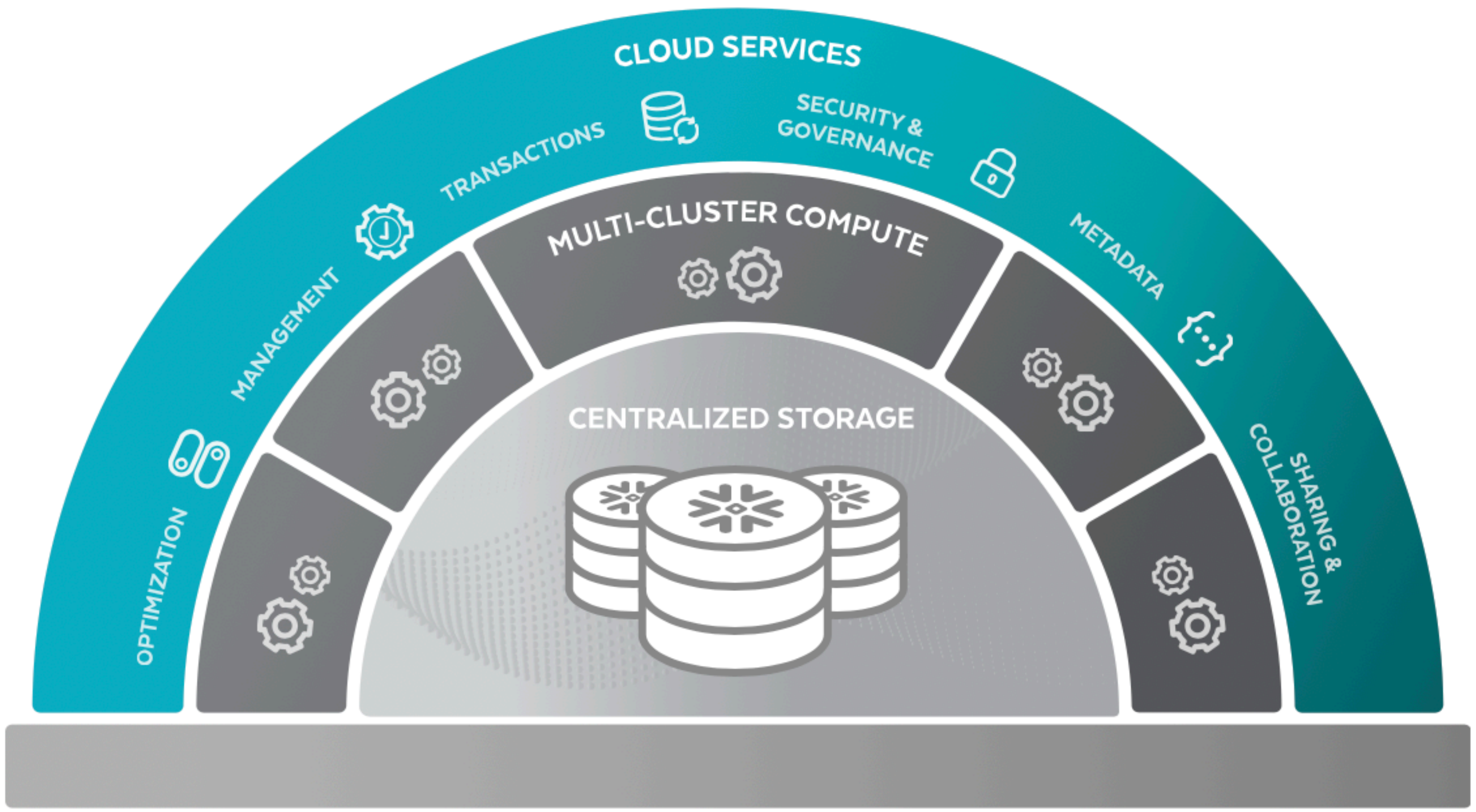


- Synapse Analytics is a data warehouse cloud service
- Can be deployed as dedicated cluster or as serverless service
- Massively Parallel Processing (MPP) SQL engine
- Integration with Apache Spark and other Azure services
- Scales to Petabyes of data

Azure Synapse Analytics cloud data warehouse

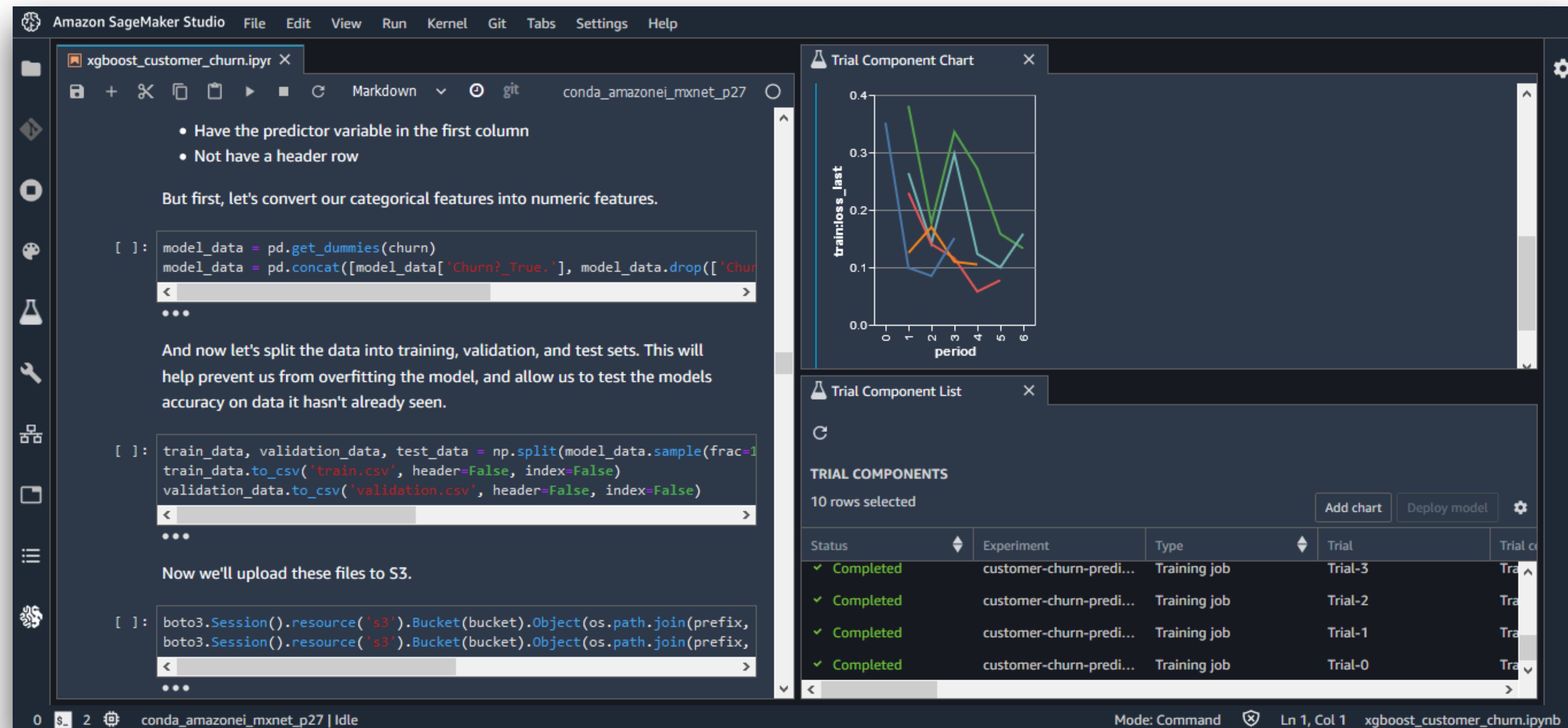


Snowflake cloud data warehouse



- Snowflake is a data warehouse cloud service
- Massively Parallel Processing (MPP) SQL engine
- Scales to Petabyes of data

Amazon SageMaker machine learning platform



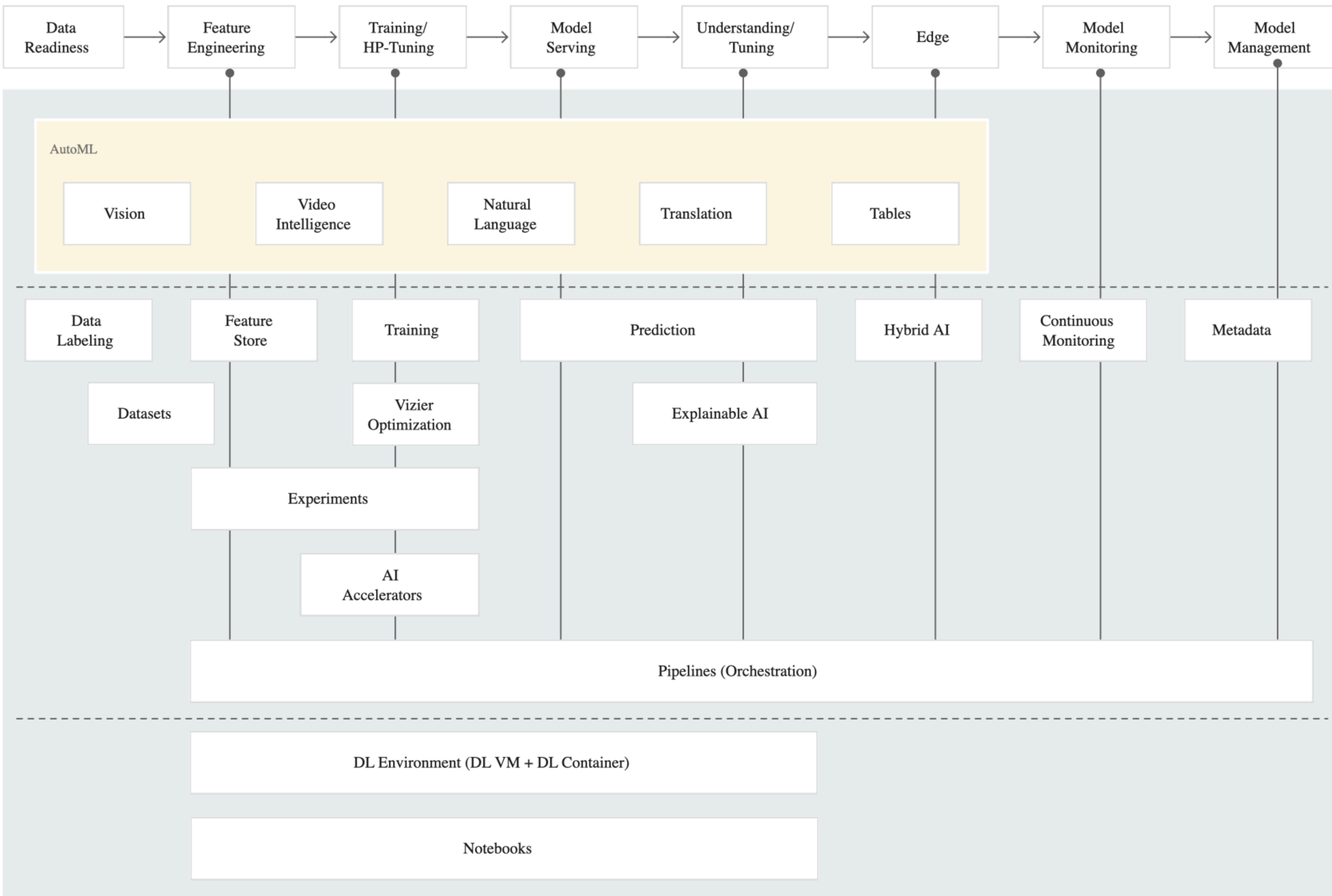
- SageMaker is a machine learning platform in the cloud
- Prepare data, build, train, and deploy machine learning models

Amazon SageMaker machine learning platform

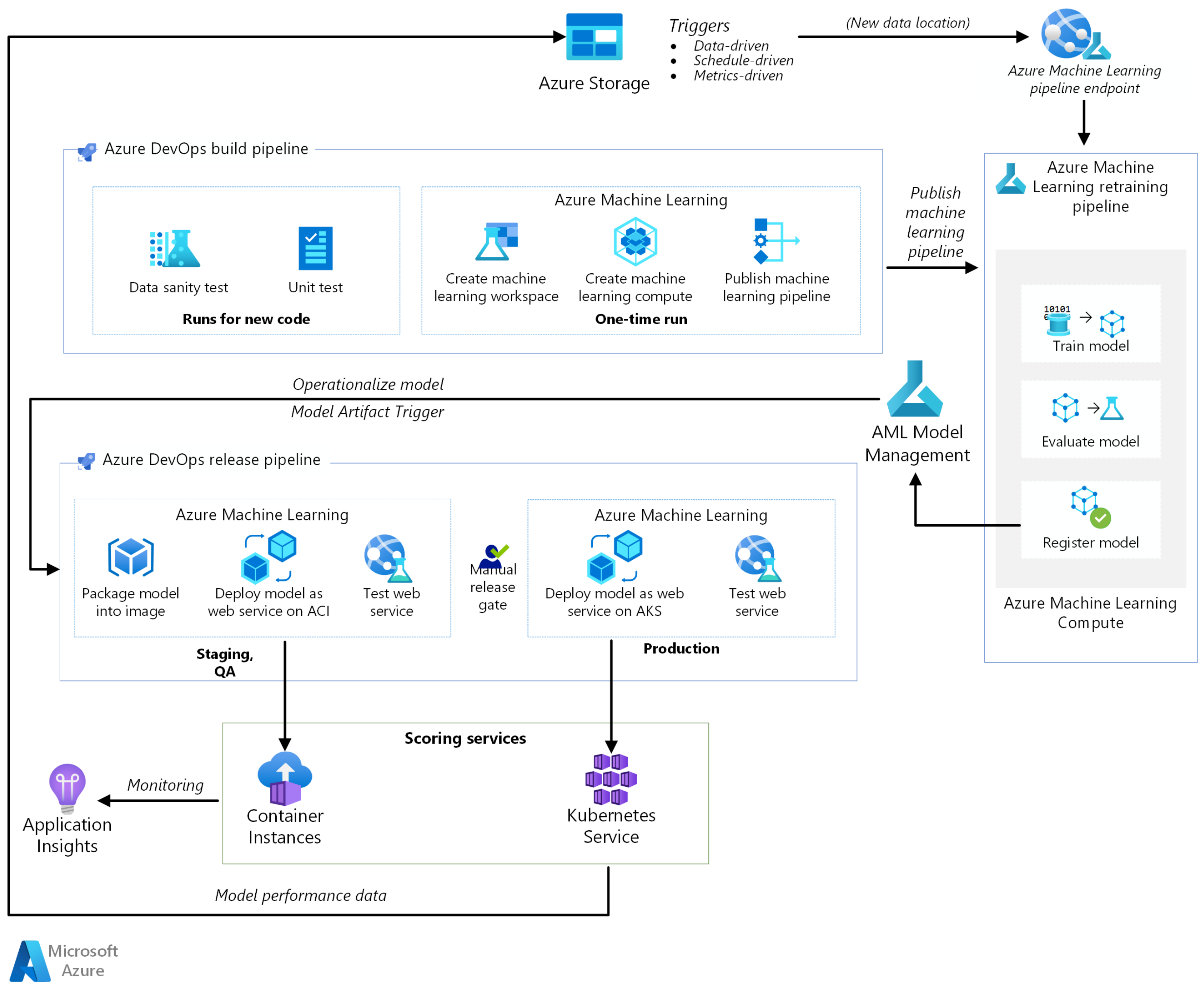
Features

- SageMaker **Notebooks** with Python, Spark and R kernels
- SageMaker **Clarify** detects statistical bias across the entire ML workflow and includes tools to help explain ML models
- SageMaker **Studio** provides a single UI for all ML development steps. You can quickly upload data, create new notebooks, train and tune models, move back and forth between steps to adjust experiments, compare results, and deploy models to production all in one place
- SageMaker **Model Cards** capture model metadata, which can be shared between developers
- SageMaker **Role Manager** provides access control for shared models
- SageMaker **Model Dashboard** gives overview of deployed models
- SageMaker **Autopilot** automatically picks the best algorithm for the data
- SageMaker **JumpStart** provides 150 pre-trained open source models from PyTorch Hub and TensorFlow Hub and allows sharing ML artefacts between developers
- SageMaker **Canvas** is a no-code service with a point-and-click interface to create ML-based predictions
- SageMaker **Pipelines** helps you create fully automated ML workflows. Also provides a model registry.
- SageMaker **DataWrangler** allows you to browse and import data (similar to Glue DataBrew) and create features
- SageMaker **Feature Store** provides a central repository for data features with low latency (milliseconds) reads and writes. Features can be stored, retrieved, discovered, and shared for easy across models and teams.
- SageMaker **Ground Truth** allows you to perform data labelling on images, text files and videos.
- SageMaker **Canvas** is a visual drag-and-drop service that allows business analysts to build ML models and generate accurate predictions without writing any code or requiring ML expertise.
- SageMaker **Experiments** helps you organize and track iterations to ML models. It automatically captures the input parameters, configurations, and results, and stores them as "experiments".
- SageMaker **Debugger** automatically captures real-time metrics during training, such as confusion matrices and learning gradients, to help improve model accuracy.
- SageMaker **Training Compiler** is a deep learning (DL) compiler that accelerates DL model training by up to 50 percent through graph- and kernel-level optimizations to use GPUs more efficiently.
- For model deployment, SageMakers offers three options
  - Real-time inference
  - Batch transform
  - Asynchronous inference

Google Vertex AI machine learning platform



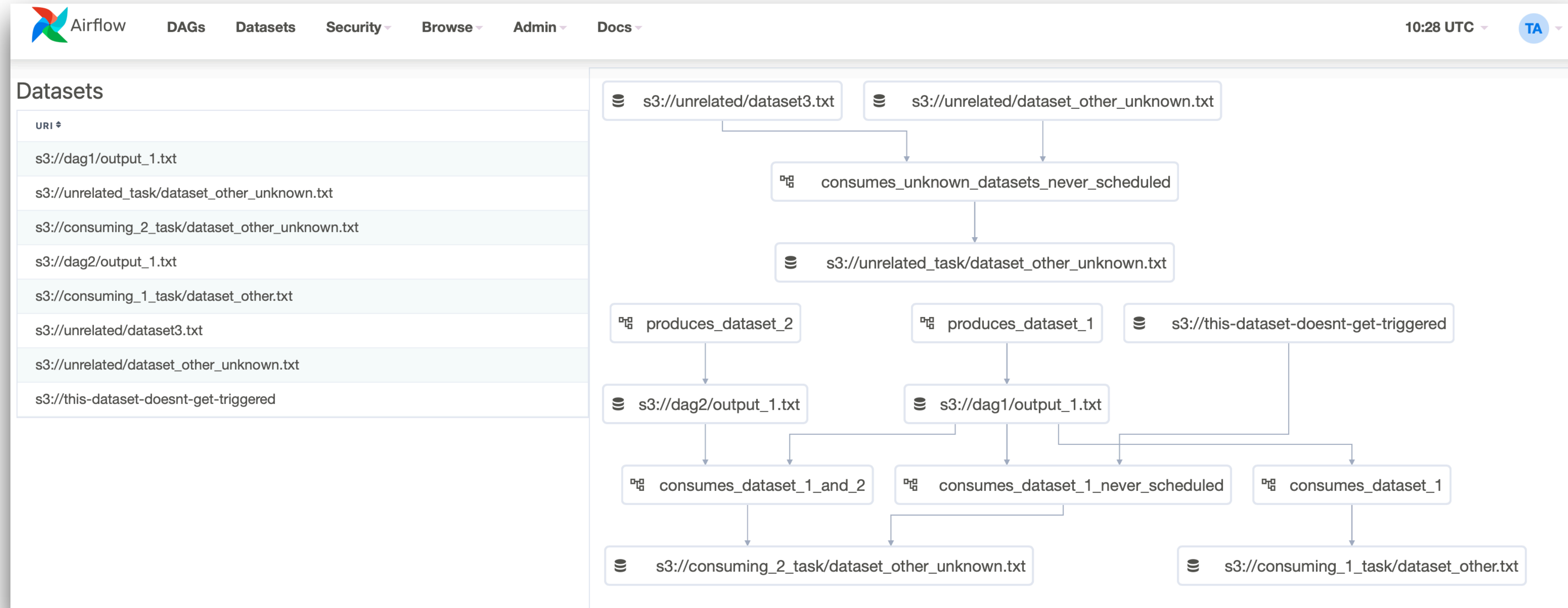
Azure Machine Learning platform



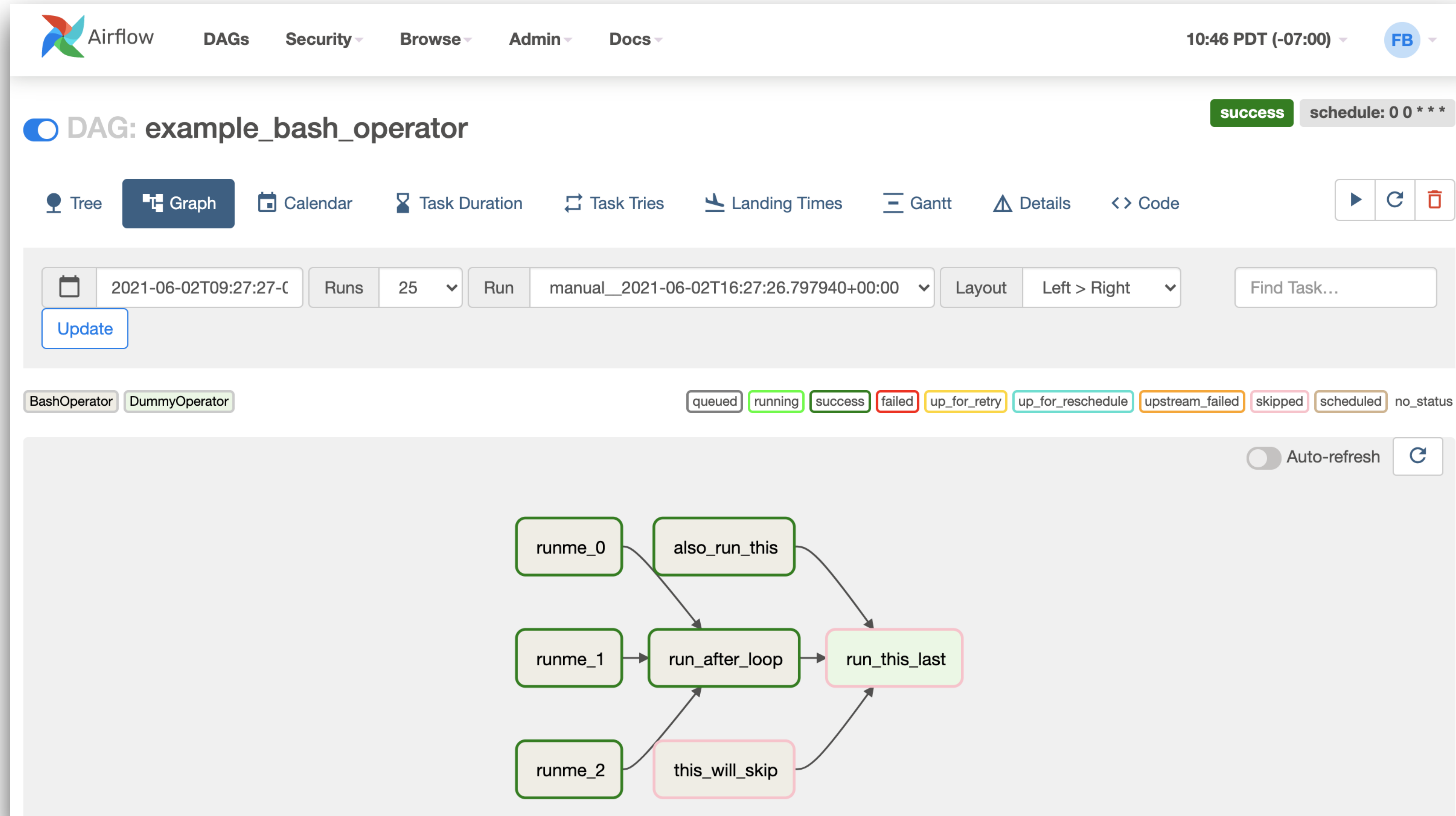
Apache Airflow workflow management software

- Initially developed at Airbnb in 2014 to manage data engineering workflows / pipelines
- 2016 Apache incubator project
- 2019 Apache top-level project
- Written in Python
- Developer defines workflow as directed acyclic graph (DAG)
  - Writes workflow dependencies in Python
  - Writes workflow tasks in Python
- Integration with popular databases, storage services, analytics engines and cloud services via pluggable *operators*.
- Cloud providers offer managed version of Airflow as cloud service
  - Google Cloud Composer
  - AWS Managed Workflows for Apache Airflow

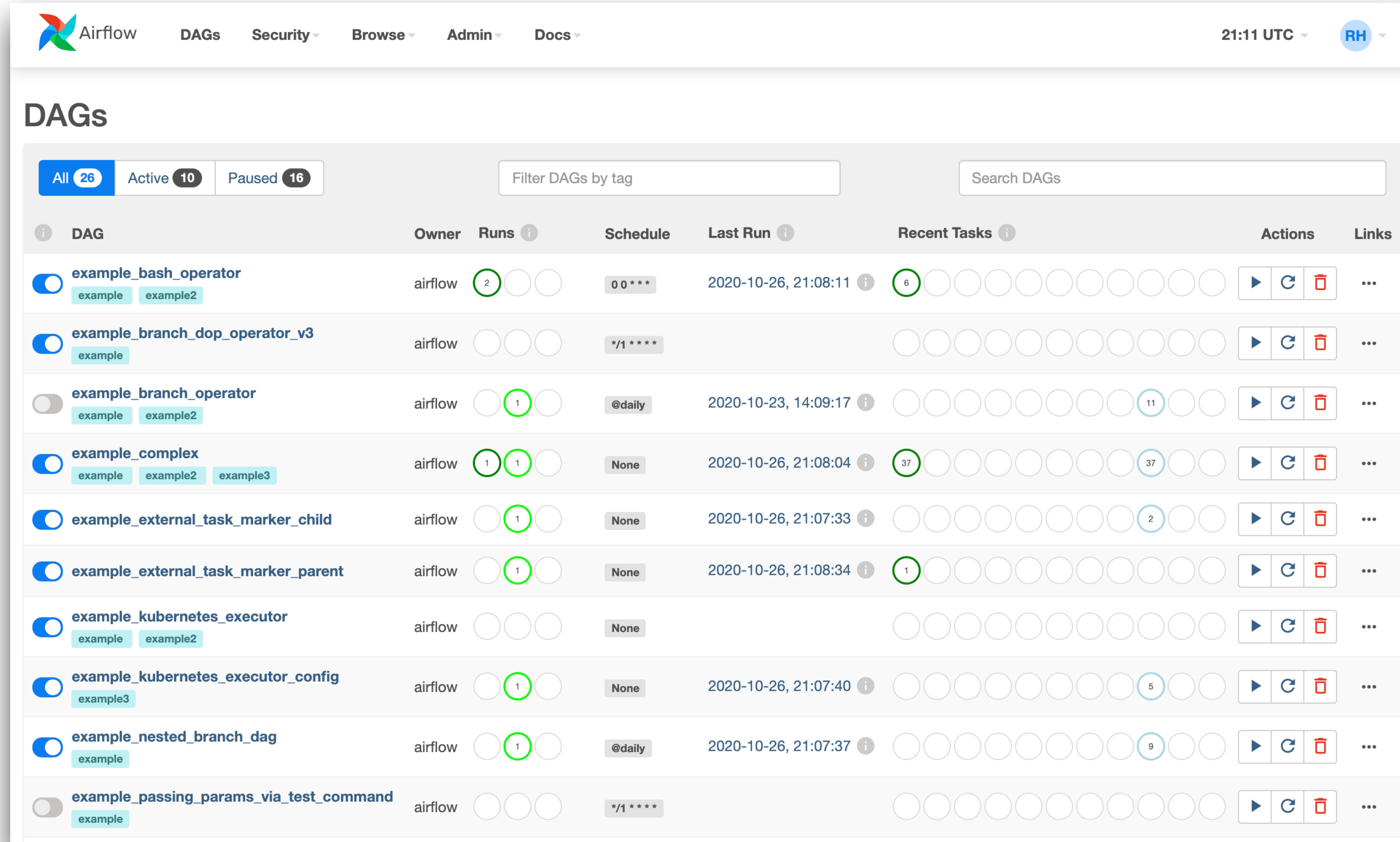
Apache Airflow workflow management software



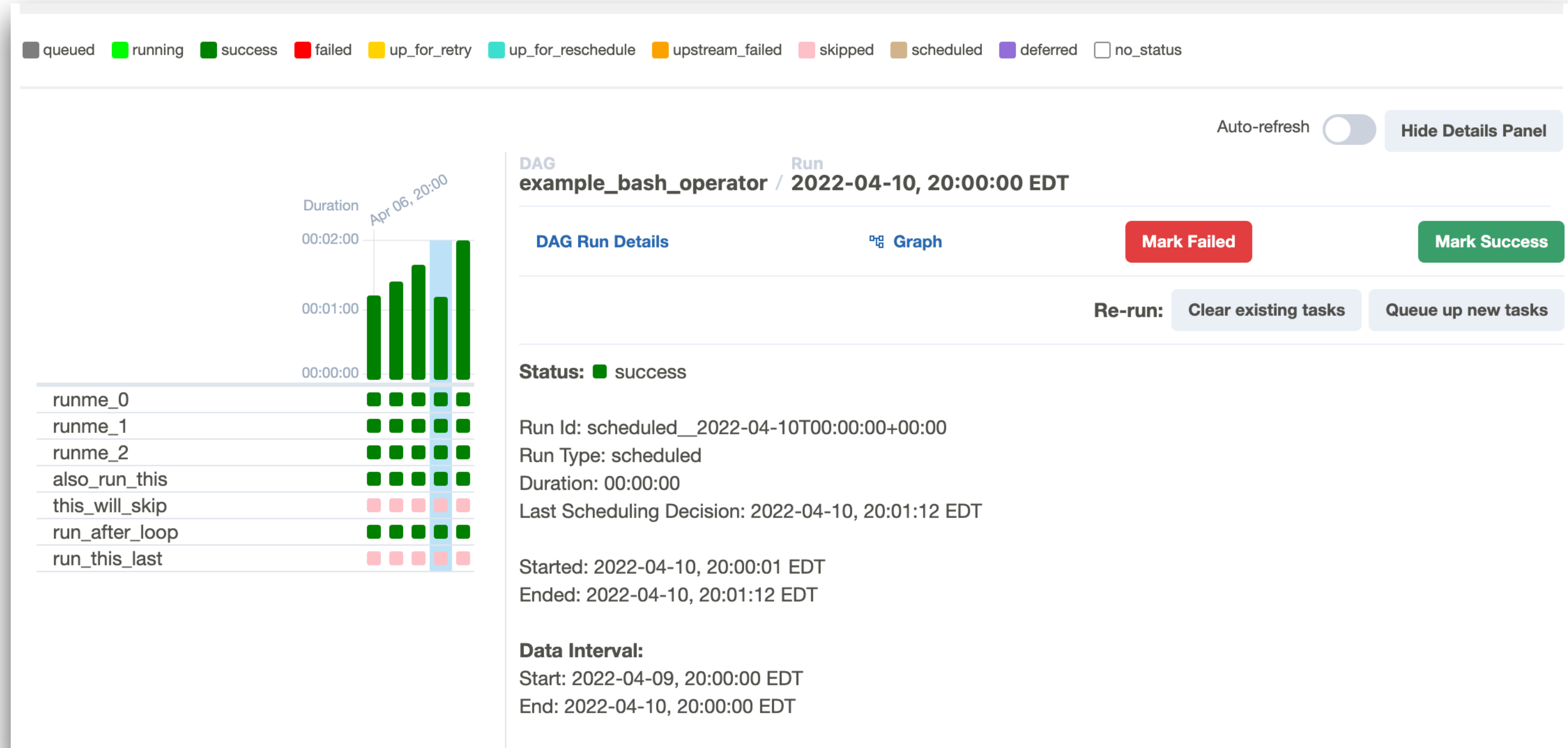
Apache Airflow workflow management software



Apache Airflow workflow management software



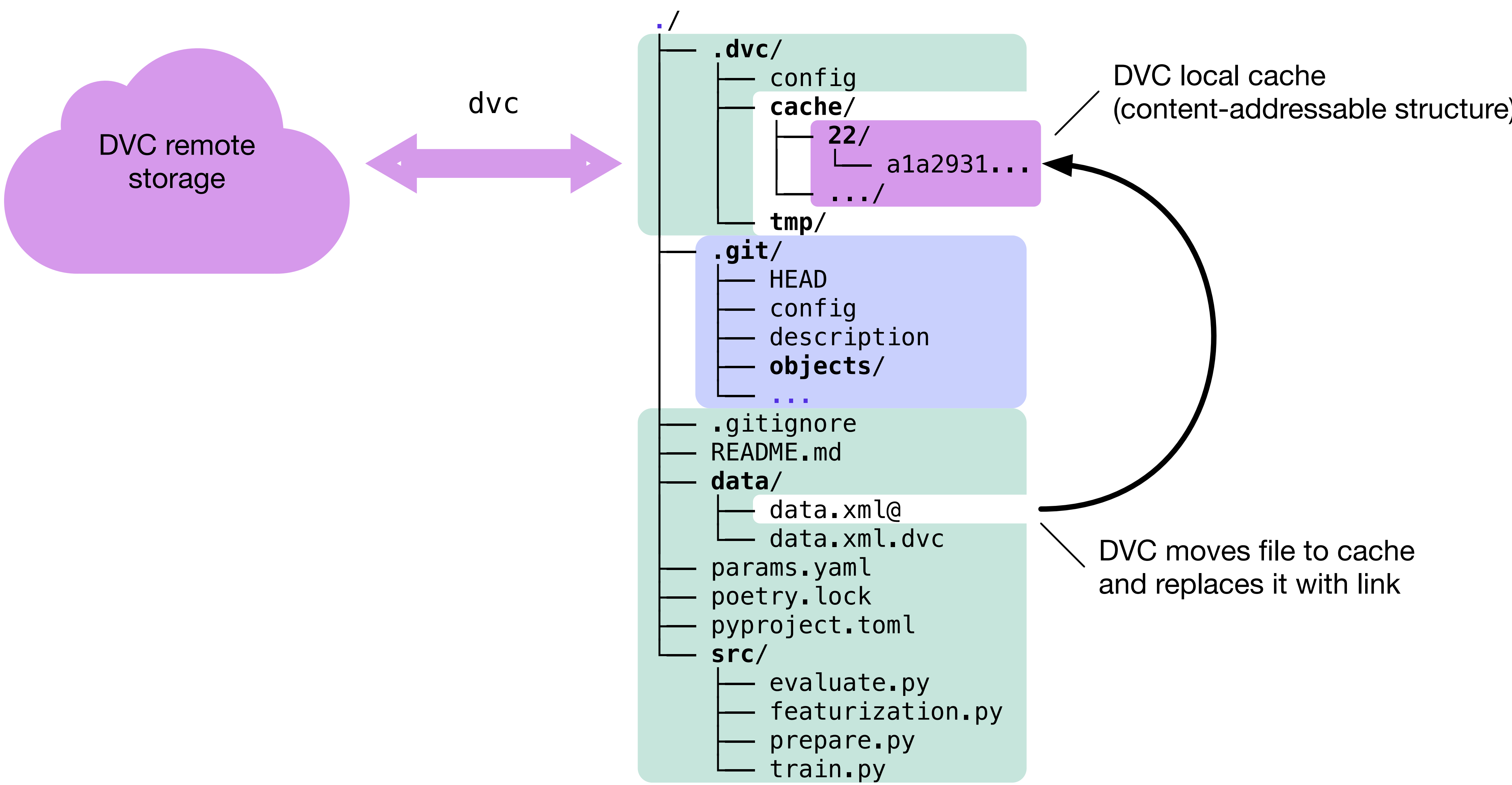
Apache Airflow workflow management software



Data Version Control (DVC)

- Open Source project developed by Iterative.ai
  - Initial release 2017-05
  - Written in Python
- Runs on top of any Git repository
  - On the local machine, code and data sets are in the same workspace
  - Data sets are not stored in Git, but only documents pointing to them
  - Data sets are stored in S3 or similar
  - On the local machine, the data is kept in a cache directory (default `.dvc/cache`). The workspace contains symbolic links to the files in the cache.
- Works as a command-line tool

Data Version Control (DVC)

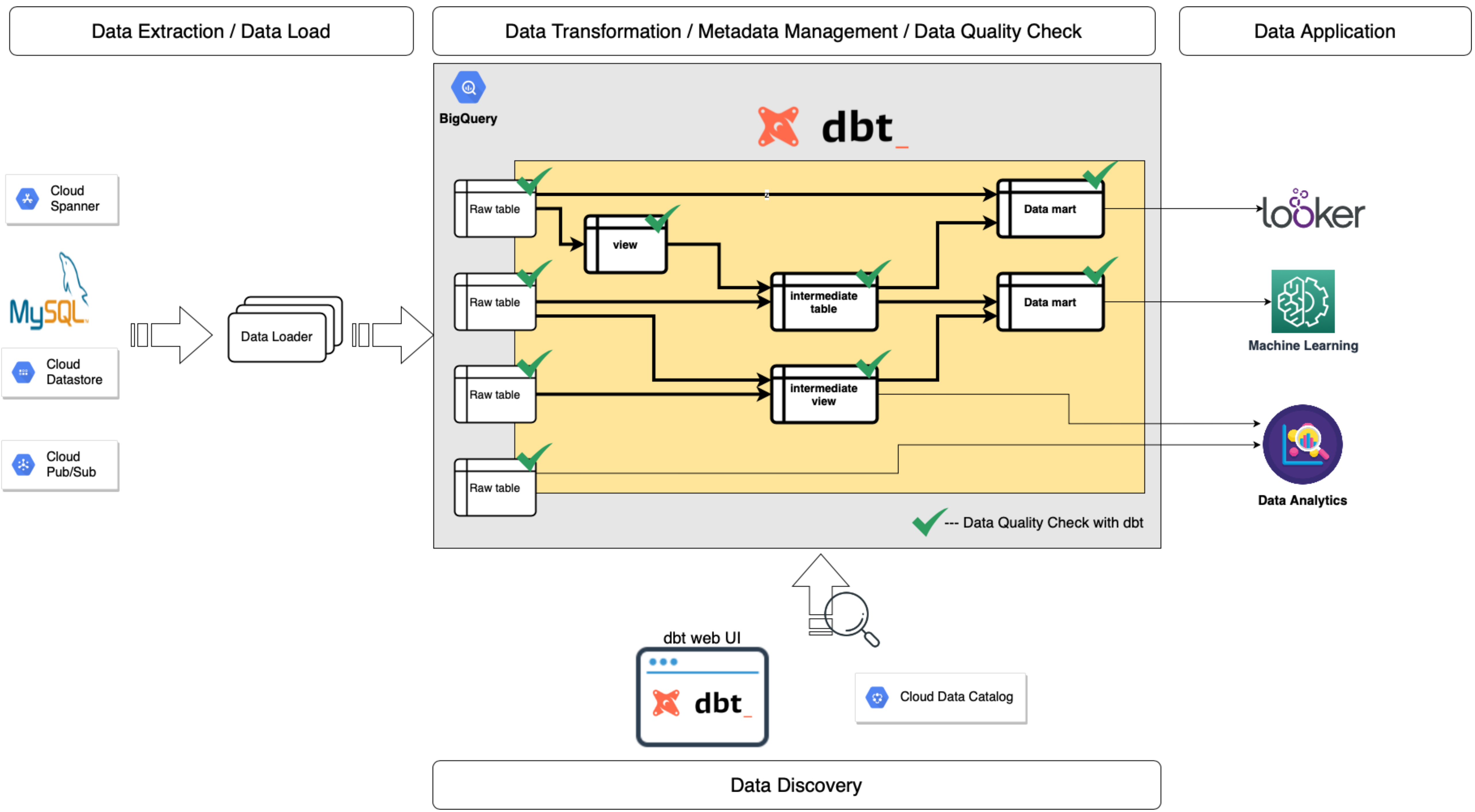


Data Build Tool (dbt)

- Tool for transforming relational data to use it in a data warehouse.
  - Open Source
  - Developed by dbt Labs
- Command-line tool to read source data, transform the data and produce data ready for consumption.
- Transformations are written as SQL SELECT statements.
  - Several transformations can be chained in a pipeline.
  - Several pipelines can share the same transformations.
- To store the data and execute the transformations, dbt connects to a database (local, on premises or in the cloud).
- The SQL code for transformations is stored in a project directory and versioned in Git.

- For "building" the final data, dbt uses the same practices as building an app from source code:
  - dbt automatically manages dependencies between transformations
  - dbt is able to automatically execute data quality tests
  - dbt enables a team to share the build pipeline

Data Build Tool (dbt)



Data Build Tool (dbt)